



# Applied ML

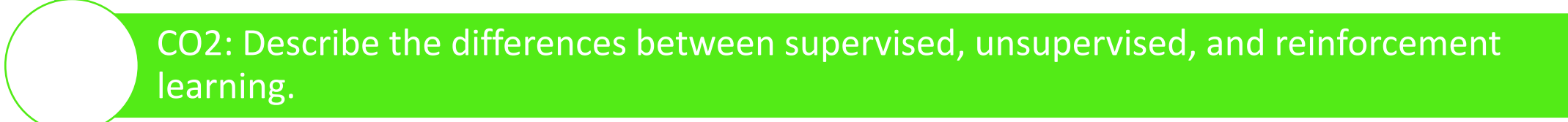
**Prerequisite(s):** Elements of AI

|                                     |                                |
|-------------------------------------|--------------------------------|
| <b>Total Units to be Covered: 6</b> | <b>Total Contact Hours: 48</b> |
|-------------------------------------|--------------------------------|

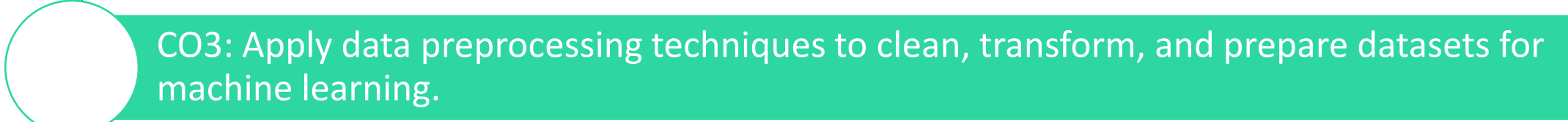
# Course Outcomes



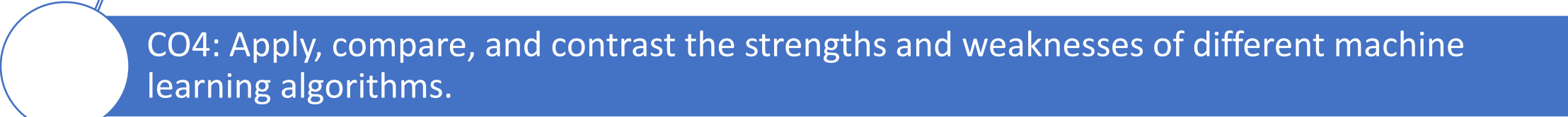
CO1: Recall and define key machine learning concepts, terminologies, and algorithms.



CO2: Describe the differences between supervised, unsupervised, and reinforcement learning.



CO3: Apply data preprocessing techniques to clean, transform, and prepare datasets for machine learning.



CO4: Apply, compare, and contrast the strengths and weaknesses of different machine learning algorithms.

# UNDERSTANDING REGRESSION LOSS FUNCTIONS

## MSE, MAE, & HUBER LOSS



# MAE vs MSE vs HUBER LOSS

Choosing the right regression loss function

## MAE

Mean Absolute Error

$$\text{MAE} = (1/n) \sum |y_i - \hat{y}_i|$$

Average absolute prediction error

### PROS

- ✓ Robust to outliers
- ✓ Same unit as the target
- ✓ Easy to interpret

### CONS

- ✗ Not differentiable at zero
- ✗ Constant gradient may slow fine-tuning

Use when data contains outliers and every error should receive equal weight.

Example: house prices with extreme values

## MSE

Mean Squared Error

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

Squares errors and heavily penalizes large mistakes

### PROS

- ✓ Smooth and differentiable
- ✓ Strong penalty for large errors
- ✓ Convenient for optimization

### CONS

- ✗ Highly sensitive to outliers
- ✗ Reported unit is squared

Use when large errors are especially costly and noise is approximately Gaussian.

Example: sensor calibration with clean data

## HUBER

Balanced robust loss

$$L\delta(e) = \begin{cases} \frac{1}{2}e^2 & \text{if } |e| \leq \delta \\ \delta(|e| - \frac{1}{2}\delta) & \text{if } |e| > \delta \end{cases}$$

where  $e = y - \hat{y}$

Quadratic near zero and linear for large errors

### PROS

- ✓ Smooth at the transition
- ✓ Less sensitive to outliers than MSE
- ✓ Stable gradients

### CONS

- ✗ Threshold  $\delta$  must be tuned
- ✗ Less immediately interpretable

Use for mostly clean data with occasional outliers.

Example: robust deep-learning regression

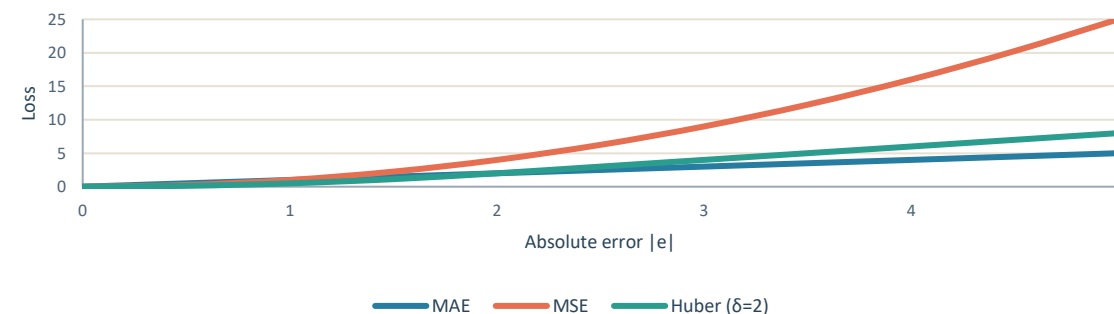
## WORKED EXAMPLE

Absolute errors:  $|e| = 1, 2, 5$  Huber threshold:  $\delta = 2$

| Error $ e $ | MAE term | MSE term | Huber term |
|-------------|----------|----------|------------|
| 1           | 1        | 1        | 0.5        |
| 2           | 2        | 4        | 2          |
| 5           | 5        | 25       | 8          |
| MEAN        | 2.67     | 10.00    | 3.50       |

The large error (5) dominates MSE. MAE and Huber limit its influence.

## LOSS vs ERROR MAGNITUDE



## MSE

(Mean Squared Error)

### Formula

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where,

$y_i$  = actual value

$\hat{y}_i$  = predicted value

$n$  = number of observations

### Numerical Example

Actual values: [3, 5, 7]

Predicted values: [2, 6, 8]

Errors: 1, -1, -1

Squared errors: 1, 1, 1

$$MSE = \frac{1 + 1 + 1}{3} = 1$$

### Use Case

- When large errors should be strongly penalized.
- Commonly used in regression problems (e.g., house price prediction).

## MAE

(Mean Absolute Error)

### Formula

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Where,

$y_i$  = actual value

$\hat{y}_i$  = predicted value

$n$  = number of observations

### Numerical Example

Actual values: [3, 5, 7]

Predicted values: [2, 6, 8]

Absolute errors: 1, 1, 1

$$MAE = \frac{1 + 1 + 1}{3} = 1$$

### Use Case

- When data has outliers and all errors should be treated more equally.
- Common in real-world datasets with noise or unusual values.

## Huber Loss

### Formula

$$L_{\delta}(e) = \begin{cases} \frac{1}{2}e^2, & |e| \leq \delta \\ \delta(|e| - \frac{1}{2}\delta), & |e| > \delta \end{cases}$$

Where,

$e$  = error ( $y - \hat{y}$ )

$\delta$  = threshold (usually 1)

### Numerical Example

Let  $\delta = 1$

1. For  $e = 2$  ( $|2| > 1$ )

$$L = 1\left(2 - \frac{1}{2}\right) = 1.5$$

2. For  $e = 0.5$  ( $|0.5| \leq 1$ )

$$L = \frac{1}{2}(0.5)^2 = 0.125$$

### Use Case

- When you want MSE for small errors but less sensitivity to outliers.
- Useful in regression with some outliers.

**Note: In Huber Loss select the threshold delta value.**

# MSE, MAE and Huber Loss

## Advantages

- Simple to calculate.
- Gives more importance to large errors.
- Widely used in regression.

## Disadvantages

- Sensitive to outliers.
- Squaring makes large errors very influential.
- Unit becomes squared.

## Scale

Same as the target squared.

**Example:** If target is in ₹ → MSE is in ₹<sup>2</sup>.

## Advantages

- Easy to understand.
- Less affected by outliers than MSE.
- Error remains in the original unit.

## Disadvantages

- Does not strongly penalize large errors.
- Absolute value is not differentiable exactly at zero.

## Scale

Same as the original target.

**Example:** If target is in ₹ → MAE is in ₹.

## Advantages

- Less sensitive to outliers than MSE.
- Smooth and differentiable.
- Combines MSE for small errors and MAE-like behavior for large errors.

## Disadvantages

- Requires choosing  $\delta$ .
- Slightly more complicated than MAE/MSE.
- Choice of  $\delta$  affects the result.

## Scale

Depends on the error region.

(Standard formula gives squared units for small errors; large-error part is linear in  $|e|$  times  $\delta$ .)



## Quick Comparison

| Loss              | Formula idea          | Outlier sensitivity | Use case                                                | Scale                                |
|-------------------|-----------------------|---------------------|---------------------------------------------------------|--------------------------------------|
| <b>MSE</b>        | $\text{Error}^2$      | High                | When large errors matter (e.g., house price prediction) | Squared units                        |
| <b>MAE</b>        | $ \text{Error} $      | Low                 | When data has outliers (e.g., delivery time)            | Original units                       |
| <b>Huber Loss</b> | MSE + MAE combination | Medium / low        | When you have outliers but want smooth optimization     | Depends on region (squared + linear) |



Easy way to remember:

**MSE** → squares the error  
→ outliers matter a lot

**MAE** → takes absolute error  
→ more robust

**Huber** → MSE for small errors +  
MAE-like for large errors

## Problems #1: Calculate MAE, MSE & Huber Loss

| Observation | Actual value $y_i$ | Predicted value $\hat{y}_i$ |
|-------------|--------------------|-----------------------------|
| 1           | 10                 | 12                          |
| 2           | 20                 | 19                          |
| 3           | 30                 | 35                          |
| 4           | 40                 | 38                          |
| 5           | 50                 | 51                          |

Same data...  
Different losses

# Regression Loss Metrics

Small changes...  
Big impact

MAE | MSE | Huber Loss

## The Data

5 observations, actual vs predicted values. Let's see how each loss sees the error differently! 😊

| Observation | Actual value $y_i$ | Predicted value $\hat{y}_i$ |
|-------------|--------------------|-----------------------------|
| 1           | 10                 | 12                          |
| 2           | 20                 | 19                          |
| 3           | 30                 | 35                          |
| 4           | 40                 | 38                          |
| 5           | 50                 | 51                          |

$y_i$  = actual  
 $\hat{y}_i$  = predicted

Same values...  
different perspectives

**Note: In Huber Loss threshold delta value=1, if not given you can assume.**

## 1 MAE

Mean Absolute Error

Takes the average of absolute differences. All errors count equally.

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Step by step:

$$\left. \begin{array}{l} |10 - 12| = 2 \\ |20 - 19| = 1 \\ |30 - 35| = 5 \\ |40 - 38| = 2 \\ |50 - 51| = 1 \end{array} \right\} \text{Sum} = 11$$

$$MAE = \frac{11}{5} = 2.2$$

Looks at the size of errors, not their squared value.



## 2 MSE

Mean Squared Error

Squares the errors, so bigger errors get more attention.

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

Step by step:

$$\left. \begin{array}{l} (10 - 12)^2 = 4 \\ (20 - 19)^2 = 1 \\ (30 - 35)^2 = 25 \\ (40 - 38)^2 = 4 \\ (50 - 51)^2 = 1 \end{array} \right\} \text{Sum} = 35$$

$$MSE = \frac{35}{5} = 7$$

Punishes large errors more.



## 3 Huber Loss

Combines the best of both worlds – like MAE for small errors and MSE for large errors.

$$L_{\delta}(\epsilon) = \begin{cases} \frac{1}{2} \epsilon^2, & |\epsilon| \leq \delta \\ \delta(|\epsilon| - \frac{1}{2} \delta), & |\epsilon| > \delta \end{cases}$$

Assume  $\delta = 1$

Loss for each error:

| Error ( $\epsilon$ ) | $ \epsilon $ | Huber Loss |
|----------------------|--------------|------------|
| -2                   | 2            | 1.5        |
| 1                    | 1            | 0.5        |
| -5                   | 5            | 4.5        |
| 2                    | 2            | 1.5        |
| -1                   | 1            | 0.5        |

$$\text{Total Huber Loss} = 1.5 + 0.5 + 4.5 + 1.5 + 0.5 = 8.5$$

$$\text{Mean Huber Loss} = \frac{8.5}{5} = 1.7$$

Less sensitive to outliers.



Final Numbers

MAE = 2.2

MSE = 7

Huber Loss = 1.7

Different losses...  
Same data...  
Different insights!

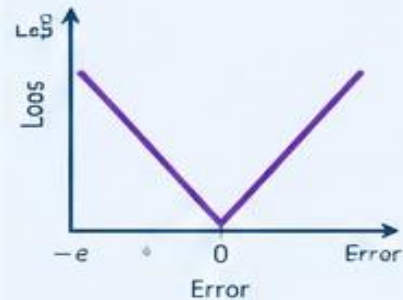


Better models...  
→ Better decisions



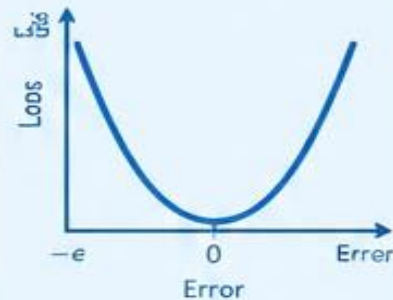
## Graphs

### MAE



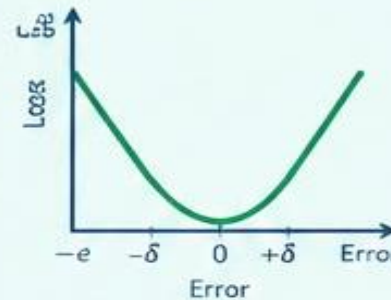
V-shaped (linear)

### MSE



Parabolic (quadratic)

### Huber Loss



Quadratic  $\rightarrow$  Linear  
(transition at  $\pm\delta$ )

## Numerical Example

Given:  $y = [10, 20, 30, 40, 50]$   
 $\hat{y} = [8, 19, 36, 38, 51]$   
 $\delta = 2$

Errors ( $e = y - \hat{y}$ ):  $[2, 1, -5, 2, 1]$

| Loss  | Calculation                   | Value |
|-------|-------------------------------|-------|
| MAE   | $(2 + 1 + 5 + 2 + 1) / 5$     | 2.2   |
| MSE   | $(4 + 1 + 25 + 4 + 1) / 5$    | 7     |
| Huber | $(2 + 0.5 + 8 + 2 + 0.5) / 5$ | 2.6   |

(For Huber:  $|e| \leq 2 \rightarrow \frac{1}{2}e^2$ ;  $|e| > 2 \rightarrow 2(|e| - 1)$ )

## Advantages & Disadvantages

### MAE

#### Advantages

- Robust to outliers
- Easy to interpret
- Linear influence of errors
- Simple to compute

#### Disadvantages

- Not differentiable at 0
- Less smooth for optimization
- Less penalty for large errors

### MSE

#### Advantages

- Simple and easy to compute
- Differentiable and smooth
- Strongly penalizes large errors
- Useful when large errors are critical

#### Disadvantages

- Highly sensitive to outliers
- Can distort model training
- Loss in squared units

### Huber Loss

#### Advantages

- Robust to outliers
- Combines MSE + MAE
- Differentiable and smooth
- Linear penalty for large errors
- Bounded gradient

#### Disadvantages

- Requires choosing  $\delta$
- Performance depends on  $\delta$
- More complex than MSE/MAE
- Still affected by large errors

## Use Cases

### MSE

- Standard regression problems
- When outliers are not a concern

### MAE

- Datasets with outliers
- When linear error influence is desired

### Huber Loss

- House price prediction
- Financial data
- Sensor data
- Regression with outliers

MSE = Square | MAE = Absolute | Huber = Square (small) + Linear (large)

## 8 Key Takeaways



- MAE → Linear penalty, robust to outliers.
- MSE → Squared penalty, more sensitive to large errors.
- Huber Loss → Combination of both (quadratic for small errors, linear for large errors), more robust than MSE.
- Use Huber Loss when you want a balance between stability and sensitivity to outliers.

$$\text{MAE} = 2.2$$

$$\text{MSE} = 7$$

$$\text{Huber Loss} = 1.7$$